

Database Technologies - Assignment 6

Use Group by clause with appropriate SQL functions to solve following queries.

Note: To solve below queries, use “sales” database

1. Write a query that counts the number of salespeople registering orders for each day. (If a salesperson has more than one order on a given day, he or she should be counted only once.).

```
KD2-Vishwajeet-96150>select odate, count(distinct snum) from orders
-> group by odate;
+-----+-----+
| odate      | count(distinct snum) |
+-----+-----+
| 1990-10-03 | 4 |
| 1990-10-04 | 3 |
+-----+-----+
2 rows in set (0.046 sec)
```

2. Write a query on the Customers table that will find the highest rating in each city. Put the output in this form:
For the city (city), the highest rating is: (rating).

```
KD2-Vishwajeet-96150>select concat('For the city ',city,' highest rating is
:', max(rating))as Rating from customers
-> group by city;
+-----+
| Rating |
+-----+
| For the city London highest rating is :100 |
| For the city Rome highest rating is :200 |
| For the city San Jose highest rating is :300 |
| For the city Berlin highest rating is :300 |
+-----+
4 rows in set (0.030 sec)
```

- 3 Write an SQL query to calculate the total order amount for each day and display the results in descending order of total orders.

```

regated column sales.orders.odate, this is incompatible with sql_mode=only_full_group_by
KD2-Vishwajeet-96150>select odate, sum(amt) as total_amt from orders
-> group by odate
-> order by sum(amt) desc;
+-----+-----+
| odate   | total_amt |
+-----+-----+
| 1990-10-04 | 16713.81 |
| 1990-10-03 | 8944.59 |
+-----+-----+
2 rows in set (0.082 sec)

KD2-Vishwajeet-96150>|

```

4. Write a query that selects the total amount in orders for each salesperson for whom this total is greater than the average amount of the order in the table.

(Note Use the average amount value directly →2565.84)

```

KD2-Vishwajeet-96150>select snum, sum(amt) from orders
-> group by snum
-> having sum(amt)> 2565.84;
+-----+-----+
| snum | sum(amt) |
+-----+-----+
| 1001 | 15382.07 |
| 1002 | 5546.15 |
+-----+-----+
2 rows in set (0.054 sec)

KD2-Vishwajeet-96150>|

```

5. Write a query that selects the highest rating in each city.

```
KD2-Vishwajeet-96150>select city, max(rating) from customers
-> group by city;
+-----+-----+
| city   | max(rating) |
+-----+-----+
| London |          100 |
| Rome   |          200 |
| San Jose |         300 |
| Berlin |         300 |
+-----+-----+
4 rows in set (0.048 sec)

KD2-Vishwajeet-96150>|
```

6. Retrieve the maximum order amount for each salesperson from the orders table, including only those orders whose value exceeds Rs. 3000.

```
KD2-Vishwajeet-96150>select snum, max(amt) from orders
-> group by snum
-> having max(amt)>3000;
+-----+-----+
| snum | max(amt) |
+-----+-----+
| 1001 | 9891.88 |
| 1002 | 5160.45 |
+-----+-----+
2 rows in set (0.036 sec)

KD2-Vishwajeet-96150>|
```

7. Write an SQL query to find the smallest order placed by every customer.

```
KD2-Vishwajeet-96150>select cnum, min(amt) from orders
-> group by cnum;
+-----+-----+
| cnum | min(amt) |
+-----+-----+
| 2008 |    18.69 |
| 2001 |   767.19 |
| 2007 |  1900.10 |
| 2003 |  5160.45 |
| 2002 |  1713.23 |
| 2004 |    75.75 |
| 2006 |  4723.00 |
+-----+-----+
7 rows in set (0.042 sec)

KD2-Vishwajeet-96150>
```